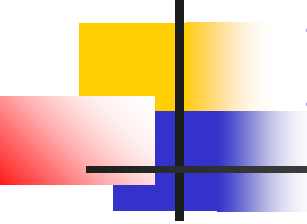


Linear Regression

Dr. Nguyen Thi Kim Ngan

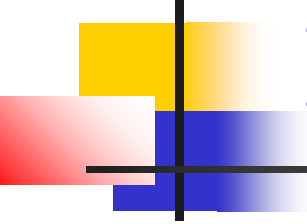


Example

Given the price of 1000 houses in a city. Each house includes the following features: size (feet²), number of bedrooms, age of house (years)

Size (feet ²)	Number of bedrooms	Age of house (years)	Price (\$1000)
2000	5	45	400
1000	3	40	200
1500	3	30	300
800	2	30	170
...	...	...	...
2500	5	20	500

How can we predict the price of another house in this city?



Example

Observation:

- The larger the house, the higher the price
- The greater the number of bedrooms, the higher the house price
- The higher the age of the house, the lower its price

=> The relationship between the features and the price of a house can be represented by a **linear function**

Example

- **Notation:**

m: number of features of a house. Each house is named a training sample

N: number of training samples

x_i : feature vector of i^{th} training sample

x_{ij} : value of j^{th} feature in x_i

y_i : label of i^{th} training sample (price of i^{th} house)

- **In this example:**

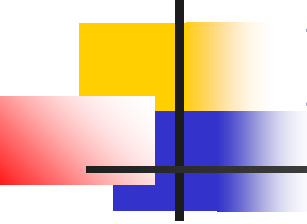
m=3, N=1000

$$x_1 = \begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \end{bmatrix} = \begin{bmatrix} 2000 \\ 5 \\ 45 \end{bmatrix} = [2000, 5, 45]^T, y_1 = 400$$

$$x_{11} = 2000, \quad x_{12} = 5, \quad x_{13} = 45$$

	Size (feet ²)	Number of bedrooms	Age of house (years)
x_1	2000	5	45
x_2	1000	3	40
x_3	1500	3	30
x_4	800	2	30
...	...	...	...
x_{1000}	2500	5	20

Price (\$1000)	
400	y_1
200	y_2
300	y_3
170	y_4
...	...
500	y_{1000}



Example

The features have different effects on the price. Suppose:

w_1 : weight of the size

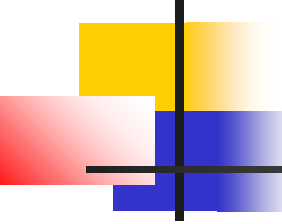
w_2 : weight of the number of bedrooms

w_3 : weight of the age of house

Notation: weight vector $w = [w_1, w_2, w_3]$

The relationship between the features of the i^{th} house $\mathbf{x}_i = [x_{i1}, x_{i2}, x_{i3}]^T$ and its predicted price $f(x_i)$ is represented by the following linear function:

$$f(\mathbf{x}_i) = x_{i1}w_1 + x_{i2}w_2 + x_{i3}w_3 = \mathbf{x}_i^T \mathbf{w}$$



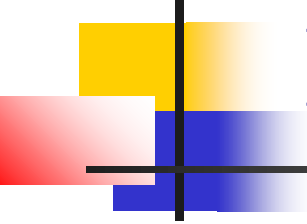
Example

The relationship between the features of the i^{th} house $\mathbf{x}_i = [x_{i1}, x_{i2}, x_{i3}]^T$ and its predicted price $f(\mathbf{x}_i)$ is represented by the following linear function:

$$f(\mathbf{x}_i) = x_{i1}w_1 + x_{i2}w_2 + x_{i3}w_3 = \mathbf{x}_i^T \mathbf{w}$$

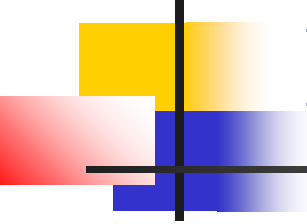
Suppose the real price of the i^{th} house is y_i , we need to find $\mathbf{w} = [w_1, w_2, w_3]$ such that the predicted price is close to the real price:

$$f(\mathbf{x}_i) = \mathbf{x}_i^T \mathbf{w} \approx y_i$$



Linear Regression

- **Input:** Training data set includes N samples. Each sample is a couple (x_i, y_i)
 - $\mathbf{x}_i$: feature vector, $\mathbf{x}_i \in \mathbb{R}^m$
 - y_i : label of x_i , $y_i \in \mathbb{R}$
- **Output:** the weight vector ($\mathbf{w}$) of the linear function $f(\mathbf{x}_i) = \mathbf{x}_i^T \mathbf{w}$ such that $f(\mathbf{x}_i) \approx y_i, \forall i=1, \dots, N$



Linear function

Given a sample $\mathbf{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \dots \\ x_{im} \end{bmatrix} \Rightarrow \mathbf{x}_i^T = [x_{i1}, x_{i2}, \dots, x_{im}]$

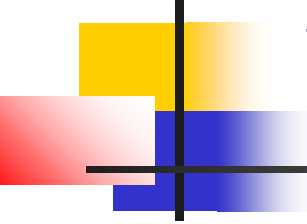
weight vector $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \dots \\ w_m \end{bmatrix}$

$f(\mathbf{x}_i)$ is the predicted label of $\mathbf{x}_i$:

$$f(\mathbf{x}_i) = x_{i1}w_1 + x_{i2}w_2 + \dots + x_{im}w_m = \mathbf{x}_i^T \mathbf{w}$$

Set:

$$\hat{y}_i = f(\mathbf{x}_i) = \mathbf{x}_i^T \mathbf{w}$$



Prediction Error

- For the predicted label is close to the real label ($\hat{y}_i \approx y_i$), the error between the predicted label and the real label must be minimum

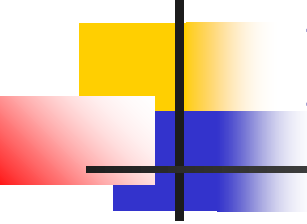
So, we need to find w such that the following expression is minimum

$$(y_i - \hat{y}_i)^2 = (y_i - \mathbf{x}_i^T w)^2$$

- Training set has N samples: $(\mathbf{x}_i, y_i)$, $i=1, 2, \dots, N$

Find w such that the mean square error (MSE) of N samples is minimum

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \mathbf{x}_i^T w)^2$$



Loss function

- Training set has N samples: $(\mathbf{x}_i, y_i)$, $i=1, 2, \dots, N$
- Find w such that the mean square error of N samples is minimum

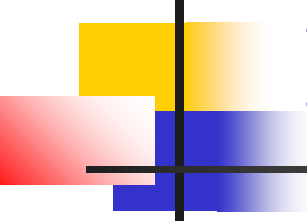
$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \mathbf{x}_i^T w)^2$$

$$L(w) = \frac{1}{2} MSE$$

$$L(w) = \frac{1}{2N} \sum_{i=1}^N (y_i - \mathbf{x}_i^T w)^2$$

=> Find w so that the loss function ($L(w)$) is minimum

$$w^* = \underset{w}{\operatorname{argmin}} L(w)$$



Loss function

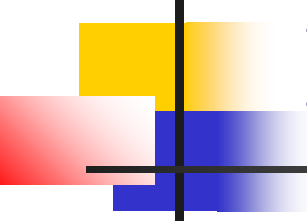
- Training set has N samples: $(\mathbf{x}_i, y_i)$, $i=1, 2, \dots, N$

$$L(w) = \frac{1}{2N} \sum_{i=1}^N (y_i - \mathbf{x}_i^T w)^2$$

- Set $X = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$, $y = [y_1, y_2, \dots, y_N]^T$

The loss function $L(w)$ is rewritten as follows:

$$L(w) = \frac{1}{2N} \sum_{i=1}^N (y_i - \mathbf{x}_i^T w)^2 = \frac{1}{2N} \left\| \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} - \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{bmatrix} w \right\|_2^2 = \frac{1}{2N} \|y - X^T w\|_2^2$$



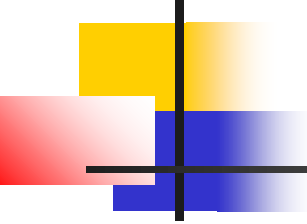
Loss function

The loss function $L(w)$ is rewritten as follows:

$$L(w) = \frac{1}{2N} \sum_{i=1}^N (y_i - \mathbf{x}_i^T w)^2 = \frac{1}{2N} \left\| \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} - \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{bmatrix} w \right\|_2^2 = \frac{1}{2N} \|y - X^T w\|_2^2$$

$L(w)$ is a quadratic function with w . The coefficient of w^2 is positive

Therefore, $L(w)$ is minimum where its derivative is 0



Optimization solution

$$L(w) = \frac{1}{2N} \|y - X^T w\|_2^2$$

- The derivative of $L(w)$ with respect to w :

$$\frac{\partial L(w)}{\partial w} = \frac{1}{N} X(X^T w - y)$$

- Solve the equation $\frac{\partial L(w)}{\partial w} = 0$

$$\frac{\partial L(w)}{\partial w} = 0 \Leftrightarrow XX^T w = Xy \Leftrightarrow \mathbf{w} = (XX^T)^\dagger Xy$$

Example

- Given a training data including 4 houses:

$$X = \begin{bmatrix} 2000 & 1000 & 1500 & 800 \\ 5 & 3 & 3 & 2 \\ 45 & 40 & 30 & 36 \end{bmatrix}$$

$$y = \begin{bmatrix} 400 \\ 200 \\ 300 \\ 170 \end{bmatrix}$$

$$w = (XX^T)^{\dagger} Xy = \begin{bmatrix} 0.207 \\ -7.818 \\ 0.497 \end{bmatrix}$$

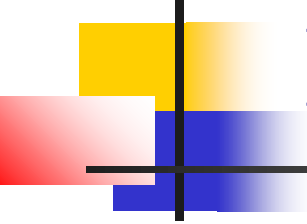
- Predict the price of a house $x = \begin{bmatrix} 1200 \\ 4 \\ 30 \end{bmatrix}$

$$f(x) = x^T w = 1200 * 0.207 + 4 * (-7.818) + 30 * 0.497 = 232.523$$

	Size (feet ²)	Number of bedrooms	Age of house (years)	Price (\$1000)	
x_1	2000	5	45	400	y_1
x_2	1000	3	40	200	y_2
x_3	1500	3	30	300	y_3
x_4	800	2	30	170	y_4

- Predict the price of a house x_1
 $f(x_1) = x_1^T w = 2000 * 0.207 + 5 * (-7.818) + 45 * 0.497 = 398.081$

$$f(x_1) \approx y_1$$



Bias trick

Adding the coefficient b makes the model slightly more flexible by removing the constraint that the linear relationship between the input and output must always pass through the origin

$$y = \mathbf{x}^T \mathbf{w} + b$$

Set $x_0=1$

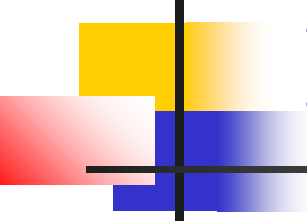
$$\bar{\mathbf{x}} = [x_0, x_1, x_2, \dots, x_N]^T$$

$$\bar{\mathbf{w}} = [b, w_1, w_2, \dots, w_N]$$

$$\bar{\mathbf{X}} = [\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2, \dots, \bar{\mathbf{x}}_N]$$

Then

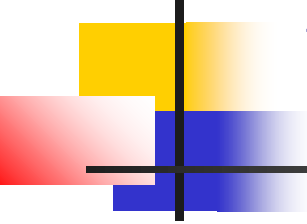
$$y = \mathbf{x}^T \mathbf{w} + b = w_1 x_1 + w_2 x_2 + \dots + w_d x_d + b x_0 = \bar{\mathbf{x}}^T \bar{\mathbf{w}}$$



Bias trick

$$y = \mathbf{x}^T \mathbf{w} + b = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d + b x_0 = \bar{\mathbf{x}}^T \bar{\mathbf{w}}$$

$$\bar{\mathbf{w}} = \operatorname{argmin}_{\mathbf{w}} \frac{1}{2N} \|\mathbf{y} - \bar{\mathbf{X}}^T \bar{\mathbf{w}}\|_2^2 = (\bar{\mathbf{X}} \bar{\mathbf{X}}^T)^\dagger \bar{\mathbf{X}} \mathbf{y}$$



Regression model performance evaluation

1. Nash–Sutcliffe efficiency (NSE)

$$\text{NSE} = 1 - \frac{\sum_{i=1}^n (H_i^{\text{mea}} - H_i^{\text{pre}})^2}{\sum_{i=1}^n (H_i^{\text{mea}} - \bar{H}^{\text{mea}})^2} \quad (16)$$

2. Coefficient of determination (R^2)

$$R^2 = \left[\frac{\sum_{i=1}^n (H_i^{\text{mea}} - \bar{H}^{\text{mea}})(H_i^{\text{pre}} - \bar{H}^{\text{pre}})}{\sqrt{\sum_{i=1}^n (H_i^{\text{mea}} - \bar{H}^{\text{mea}})^2 \sum_{i=1}^n (H_i^{\text{pre}} - \bar{H}^{\text{pre}})^2}} \right]^2 \quad (17)$$

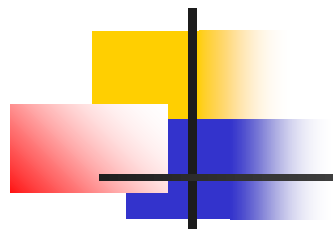
3. Mean absolute error (MAE)

$$\text{MAE} = \frac{\sum_{i=1}^n |H_i^{\text{mea}} - H_i^{\text{pre}}|}{n} \quad (18)$$

4. Root mean square error (RMSE)

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (H_i^{\text{mea}} - H_i^{\text{pre}})^2}{n}} \quad (19)$$

where H_i^{mea} and H_i^{pre} are, respectively, the measured and predicted river stages and $\bar{H}^{\text{mea}}$ and $\bar{H}^{\text{pre}}$ are, respectively, the mean of the measured and predicted river stages.



THANK YOU FOR YOUR ATTENTION!